

Hawaii Cropland Data Layers (HCDL) V2.0

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Abstract:

The Hawaii Cropland Data Layer (HCDL) is the first-ever raster, geo-referenced, crop-specific land cover data layer for Hawaii. Unlike the 10-meter resolution CDL for CONUS or HCDL V1.x, which utilized 10-day image composites of surface reflectance data across multiple spectral bands and NDVI, crop classification for HCDL 2023 and 2024 V2.0 leveraged the 64-dimensional geospatial "embeddings" from the Satellite Embedding dataset developed by Google DeepMind's new geospatial AI model, AlphaEarth Foundations. This dataset was derived from multiple datastreams, including optical, radar, LiDAR, and other sources.

Ground-reference data for crop type classification was developed based on satellite/aerial imagery interpretation and field surveys conducted by the University of Hawaii at Manoa from December 2023 through October 2024. Fieldwork covered the islands of Kauai, Oahu, Maui, Molokai, and the Big Island. HCDL 2024 was derived from 2024 embeddings using the SmileRandomForest classifier in Google Earth Engine, based on the 2024 ground-reference data. Since the embedding space is consistent across years, it can be used for condition change detection. Due to the limited ground-reference data available in 2023, HCDL 2023 was derived from 2023 embeddings solely based on the trained 2024 models, thus reducing uncertainties caused by the subjective human-involved training data labeling process.

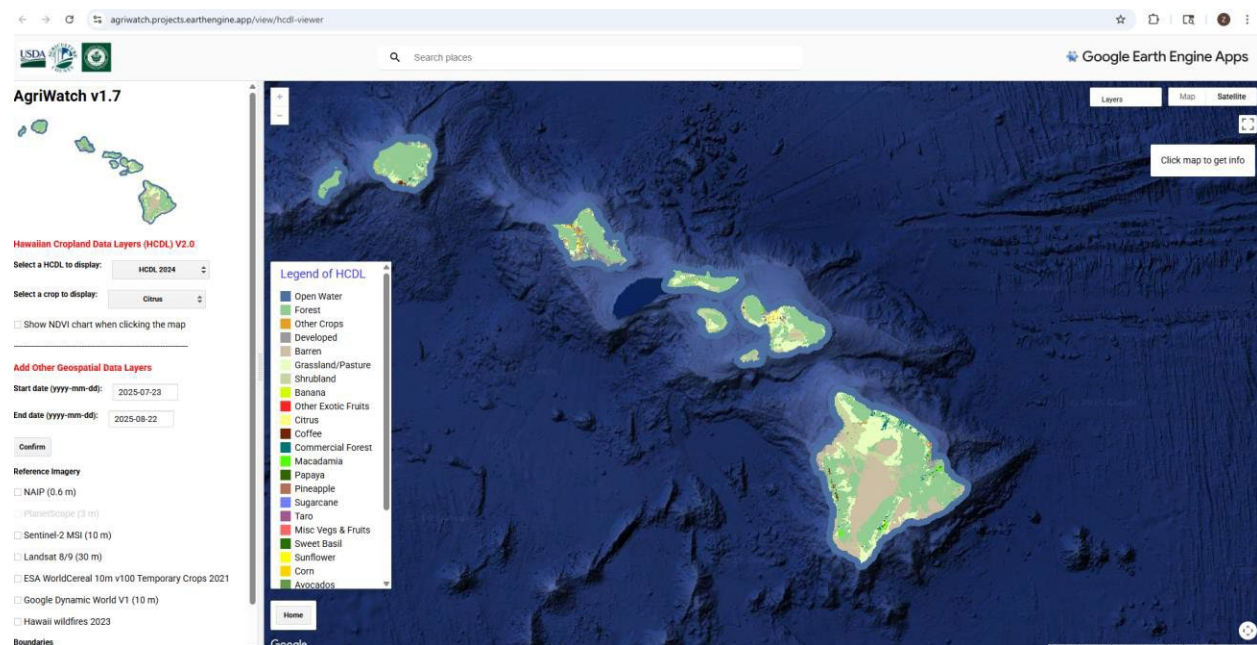


Figure 1. Hawaii Cropland Data Layers (HCDL) displayed by AgriWatch V1.7
(<https://agriwatch.projects.earthengine.app/view/hcdl-viewer>)

HCDL aims to map major, specialty, and culturally significant crops in Hawaii, as shown in Table 1. The "Other Exotic Fruits" category encompasses a variety of tropical fruits such as mango, litchi, longan, cacao, dragon fruit, jackfruit, breadfruit, and rambutan. The " Misc Veggies & Fruits " category includes tomato, pumpkin, turmeric, mustard, eggplant and others.

Table 1. HCDL Legend

HCDL Class Value	CDL Class Value	Class Name	Color Code
1	111	Open Water	#4D70A3
2	63	Forest	#93CC93
5	106	Other Crops*	#E3A12B
7	82	Developed	#9A9A9A
8	131	Barren	#CCBFA3
11	176	Grassland/Pasture	#E8FFBF
12	152	Shrubland	#C6D69E
14	93	Banana*	#D1FF00
17	94	Other Exotic Fruits*	#FF2626
19	72	Citrus	#FFFF7E
21	95	Coffee*	#702600
22	99	Commercial Forest*	#007777
34	96	Macadamia*	#54FF00
40	97	Papaya*	#336600
42	98	Pineapple*	#B5705B
48	45	Sugarcane	#757EFF
52	100	Taro*	#A05989
56	47	Misc Veggies & Fruits	#FF6666
57	101	Sweet Basil*	#267000
58	6	Sunflower	#FFFF00
59	1	Corn	#FFD300
60	215	Avocados	#66994D
62	102	Coconut*	#999900
64	103	Plumeria*	#FF6600
65	104	Noni Fruit*	#FF3399
66	105	Dracaena*	#DFB9D1
67	46	Sweet Potatoes	#732600

"Red" indicates classes that are not available in CONUS CDL

HCDL can be viewed using following web platforms including

- AgriWatch (<https://agriwatch.projects.earthengine.app/view/hcdl-viewer>)
- CroplandCROS (<https://croplandcros.scinet.usda.gov/>)
- CropScape (<https://nassgeodata.gmu.edu/CropScape/>)

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Citations:

Brown, C. F., Kazmierski, M. R., Pasquarella, V J., Rucklidge, W. J., Samsikova, M., Zhang, C., Shelhamer, E., Lahera, E., Wiles, O., Ilyushchenko, S., Gorelick, N., Zhang, L. L., Alj, S., Schechter, E., Askay, S., Guinan, O., Moore, R., Boukouvalas, A., & Kohli, P., 2025, AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data. arXiv preprint arXiv.2507.22291. [doi:10.48550/arXiv.2507.22291](https://doi.org/10.48550/arXiv.2507.22291)

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